

LARGE ERDŐS–STRAUS WITNESSES: A CUBIC Ω -RESULT FOR THE MULTIPLIER PARAMETER

ANONYMOUS

ABSTRACT. For a prime p let $W(p)$ be the least modulus $M \equiv 3 \pmod{4}$ of a “multiplier” congruence that forces a solution of $4/p = 1/x + 1/y + 1/z$, and let $ck_{\min}(p)$ be the least product ck of a positive Type I slice. Both are expected to be at most a power of $\log p$ for almost all p , and are known to exceed a constant times $\log p$ for infinitely many p . We prove that $W(p) \geq (\log p)^3 \exp(-C \log \log p / \log \log \log p)$ for infinitely many primes p in Mordell’s hard classes. The class of one is imposed modulo the primes up to $T^{1/3+o(1)}$, so that the residual congruence system consists of single-prime and two-prime events. We build a pointwise minorant of its void indicator by inclusion–exclusion truncated by the number of primes in the support, not by the number of events; its error is controlled by a pseudoforest bound for an exponential moment, after a cheap quarantine of high-degree “hub” residues. The minorant is transferred to primes by the Linnik-range prime number theorem of Thorner and Zaman, whose relative error term absorbs a possible Siegel zero. As a warm-up we give the simpler prime-local construction, which reaches exponent 2 and cannot go further. For the slice parameter we prove that congruence methods certify exactly the least quadratic non-residue, giving $ck_{\min}(p) \gg \log p \log \log \log p$ infinitely often (via Lau–Wu, after Graham–Ringrose). On the Haar side we prove $\log(1/\delta^*(T)) \leq T^{o(1)}$ for the avoider density (polylogarithmic modulo Elsholtz–Tao), identify the single-prime atoms with points of Elsholtz–Tao’s Type I variety, and show that an individual Type I bound n^η would give the Haar exponent $\eta/(1+\eta)$.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes, it has been verified for $n \leq 10^{18}$ ([11]; [14] for 10^{17} ; see [13], §6), and Mordell’s identities settle every prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [10] (cf. also [12]); we call such primes *hard*. On the exceptional-set side, Vaughan [16] proved that the number $E(N)$ of $n \leq N$ without a representation satisfies $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$; see [13] for a uniform modern account and for many exceptions to the analogous statement with

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large numerators, and [4] for the average number of solutions, which over primes $p \leq N$ is $(\log N)^3$ up to a factor $\log \log N$.

This note is about the *pointwise size* of the smallest witness. Two natural witness parameters are the following.

The multiplier parameter. For $M \equiv 3 \pmod{4}$ put $A_M = (M + 1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod{M}$, then $s = (nv + u)/M$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [16, 4]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with $W(n) = +\infty$ if no such M exists. Thus $W(p) \leq T$ supplies an explicit representation with a modulus at most T .

The slice parameter. A Type I solution with parameters (a, b, c, k) satisfies $p(a + b) = k(4abc - 1)$ and gives $4/p = 1/(ack) + 1/(bck) + 1/(pabc)$. For an admissible pair (c, k) (see §12) let $M_{c,k}(p)$ be the number of divisors of $p^2 + 4ck^2$ lying in the class $-p \pmod{4ck}$; the slice is positive iff $M_{c,k}(p) > 0$ (see §12). Put $ck_{\min}(p) = \min\{ck : M_{c,k}(p) > 0, \text{sf}(c) \notin \{1, 2, 3, 6\}\}$ (the cores 1, 2, 3, 6 are excluded because they are deterministic on hard primes).

Both parameters are expected to be small for almost all primes. For W , tails of the form $\#\{p \leq N : W(p) > T\} \ll N \exp\{-c(\log T)^2 \log \log T\}$ (in the range $\log N \gg (\log T)^3 \log \log T$) are obtained in companion work whose assembly is still under internal review; we do not use them. In the other direction, the class of one (every prime $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$) has $W(p) > T$ together with Linnik's theorem shows $W(p) \gg \log p$ infinitely often. With the smaller modulus $\text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$ and Chang's least-prime bound for smooth moduli [2], the constant becomes $5/8$ (§11). The same is true for $ck_{\min}$ via genus characters. Whether $W(p)/\log p$ is unbounded was open.

Theorem 1.1 (Proved modulo Theorem 4.1; effective). *There is an absolute constant C such that for infinitely many hard primes p*

$$W(p) \geq (\log p)^3 \exp\left(-C \frac{\log \log p}{\log \log \log p}\right).$$

More precisely, for every sufficiently large T there is a prime $p \equiv 1 \pmod{840}$ with $W(p) > T$ and $\log p \leq T^{1/3} \exp(O(\log T / \log \log T))$.

In particular the statement “ $W(p) \leq (\log p)^A$ for all large p ” is false for every $A < 3$. The cited input (Theorem 4.1) is Thorner and Zaman's prime number theorem in the Linnik range [15], including the McCurley zero-free region statement quoted there. No hypothesis on Siegel zeros is needed: the exceptional zero is absorbed by a uniqueness lemma across the family of moduli (Lemma 4.2) and a two-case analysis (Theorem 5.1).

The proof has two layers. The analytic layer (Theorem 5.1) turns any congruence minorant of $\mathbf{1}[W(n) > T]$ with controlled modulus, ℓ^1 -mass and twisted means into a prime. The combinatorial layer produces such minorants. The simplest one is *prime-local*: the class of one modulo all primes up to $T^{1/2+o(1)}$ leaves a single forbidden set at each larger prime, and a Brun minorant applies; this gives exponent 2 (Theorem 6.1), which is the limit of every prime-local design (Proposition 11.3). To go further we quarantine only the primes up to $T^{1/3+o(1)}$; then moduli with two free prime factors produce two-prime events, and these cluster at “hub” residues on which many events fire together, so that the usual Bonferroni truncation by the number of events fails (Proposition 11.6). We truncate instead by the number of *primes* in the support (§7), control the error by a pseudoforest bound for $E \prod_\ell (1 + za_\ell)$ after quarantining high-degree hub residues (§8), and verify the twist condition by a conditional form of the local lemma (§9). Exponent 3 matches the Haar (profinite) bound of Theorem 1.3; an iterated quarantine improves the Haar side much further (Theorem 13.8), while below $\theta = 1/3$ the prime side is limited by pair codegrees.

For the slice parameter the picture is different: the only congruence mechanism is the genus character, and it certifies exactly the least quadratic non-residue n_p .

Theorem 1.2 (Proved; (b) modulo [8, Prop. 5.1]).

- (a) (Proved.) $ck_{\min}(p) \geq n_p$ for every prime $p \equiv 1 \pmod{24}$, and every fixed-period congruence minorant for $\{ck_{\min}(p) > T\}$ is supported, among large primes, on primes with $n_p > T$ (Corollary 12.5).
- (b) (Proved modulo [8, Prop. 5.1].) $ck_{\min}(p) \gg \log p \cdot \log \log \log p$ for infinitely many hard primes.

Thus a congruence proof of $ck_{\min}(p) > (\log p)^{1+\delta}$ infinitely often would improve the best known unconditional (Graham–Ringrose) and GRH-conditional (Montgomery) Ω -results for n_p , and would exceed the random-model prediction $\max_{p \leq x} n_p \asymp \log x \log \log x$.

Finally, on the Haar (profinite) side, let $\delta^*(T)$ be the Haar measure of the set of $n \in \widehat{\mathbb{Z}}^\times$ with $n \equiv 1 \pmod{24}$ and $W(n) > T$, normalised within the class $1 \pmod{24}$. The class of one gives $\log(1/\delta^*(T)) \leq (2/3 + o(1))T$, and a crude prime-local bound gives $T^{1/2+o(1)}$.

Theorem 1.3 (Proved). $\log(1/\delta^*(T)) \leq T^{1/3+o(1)}$.

We do not pursue heuristics about the true size of $\delta^*(T)$ here.

Status conventions. Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Proved modulo the proof of X* means that we use a bound established inside the proof of X, as stated there. *Cited* marks external statements. *Classical* marks a standard result for which we include

a short proof from a standard reference. *Proved implication* means that the implication from an explicitly stated, open hypothesis is proved in full. *Assessment* marks heuristic discussion that is not a theorem.

2. THE ATOMS OF THE MULTIPLIER PROBLEM

For $M \equiv 3 \pmod{4}$ put $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$.

Lemma 2.1 (Proved). $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$. Consequently, for every integer n ,

$$W(n) > T \iff n \bmod M \notin \mathcal{R}(M) \text{ for every } M \leq T, M \equiv 3 \pmod{4}.$$

Proof. Since $4A_M \equiv 1 \pmod{M}$ and $\gcd(A_M, M) = 1$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides A_M^2 . Conversely let $D \mid A_M^2$. For each prime q with $q^a \parallel A_M$ and $q^d \parallel D$ ($d \leq 2a$) put $u_q = \max(0, d-a)$, $w_q = d-2u_q$, $v_q = a-u_q-w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. $\square$

Lemma 2.2 (Class of one; proved). $1 \notin \mathcal{R}(M)$ for every $M \equiv 3 \pmod{4}$.

Proof. If $-4D \equiv 1$ then $D \equiv -A_M \pmod{M}$. The map $D \mapsto A_M^2/D$ preserves this congruence, so we may assume $D \leq A_M$; then $0 < D + A_M \leq 2A_M < M$ while $M \mid D + A_M$. $\square$

3. THE PRIME-LOCAL RESIDUAL SYSTEM

Fix T large and put $\mathcal{L} = \log T$,

$$y = \sqrt{T} \exp(3\mathcal{L}/\log \mathcal{L}),$$

$$Q = \text{lcm}(24, \ell^{e_\ell} : \ell \leq y \text{ prime}, \ell^{e_\ell} \leq T \text{ maximal}),$$

and let $\mathcal{U}$ be the set of primes in $(y, T]$. Then $\log Q \leq 2y$ for large T (only $\ell \leq \sqrt{T}$ have $e_\ell \geq 2$), and $840 \mid Q$. For $\ell \in \mathcal{U}$ let

$$F_\ell = \{-4D \bmod \ell : m \leq T/\ell, m\ell \equiv 3 \pmod{4}, D \mid A_{m\ell}^2, m \mid 4D + 1\},$$

$$g(\ell) = \frac{|F_\ell|}{\ell - 1}, \quad S = \sum_{\ell \in \mathcal{U}} g(\ell).$$

All elements of F_ℓ are units.

Lemma 3.1 (Prime-local reduction; proved). *If $n \equiv 1 \pmod{Q}$ and $n \bmod \ell \notin F_\ell$ for all $\ell \in \mathcal{U}$, then $W(n) > T$.*

Proof. Let $M \leq T$, $M \equiv 3 \pmod{4}$. If every prime factor of M is at most y , then $M \mid Q$, so $n \equiv 1 \pmod{M}$ and Lemma 2.2 applies. Otherwise let $\ell > y$ divide M ; since $\ell^2 > T$ we have $\ell \parallel M$, and $m = M/\ell < T/y \leq y$, so $m \mid Q$ and $n \equiv 1 \pmod{m}$. If $n \equiv -4D \pmod{M}$ with $D \mid A_M^2$, then $m \mid 4D + 1$ and $n \bmod \ell \in F_\ell$. Now use Lemma 2.1. $\square$

Lemma 3.2 (Local sizes; proved). $g(\ell) \leq 2T\tau^*(T)^2/y^2$, where $\tau^*(T) = \max_{n \leq T} \tau(n)$. In particular $\max_\ell g(\ell) \leq 1/16$ for large T .

Proof. For each $m \leq T/\ell$ there are at most $\tau(A^2) \leq \tau(A)^2$ choices of D , and $1/(\ell-1) \leq 2/\ell$. Since $\log \tau^*(T) \leq (\log 2 + o(1))\mathcal{L}/\log \mathcal{L}$ [7, Thm. 317], the bound is $\leq 2 \exp((2 \log 2 - 6 + o(1))\mathcal{L}/\log \mathcal{L})$. $\square$

The next lemma is the key arithmetic input: the congruence $m \mid 4D + 1$ saves a factor m .

Lemma 3.3 (Mass bound; proved). *With $X = (T + 1)/4$,*

$$\begin{aligned} S &\leq \frac{4}{3}(3 + \log X) \sum_{\substack{sr^2 \leq X \\ s \text{ squarefree}}} \frac{\tau(4sr^2 + 1)}{sr} \\ &\leq 2(3 + \mathcal{L})(1 + \mathcal{L})^2 \tau^*(T + 2) = \exp\left(O\left(\frac{\log T}{\log \log T}\right)\right). \end{aligned}$$

Proof. Clearly $S \leq \sum 1/(\ell-1)$ over triples (m, ℓ, D) as in the definition of F_ℓ ; put $A = A_{m\ell}$. Since $4A \equiv 1$ and $4D \equiv -1 \pmod{m}$, we have $4A^2D^{-1} \equiv -1 \pmod{m}$, so $D \mapsto A^2/D$ preserves $m \mid 4D + 1$ and exchanges $D > A$ with $D < A$; hence it suffices to count $D \leq A$ and double. Write $D = sr^2$ with s squarefree. Then $D \mid A^2$ iff $sr \mid A$; write $A = srk$, so that $D \leq A$ iff $k \geq r$. The map $(m, \ell, D) \mapsto (m, s, r, k)$ is injective, and $1/(\ell-1) \leq 2/\ell = 2m/(4A-1) \leq 2m/(3A)$.

From $m \mid 4sr^2 + 1$ and $m \mid m\ell = 4srk - 1$ we get $m \mid 4sr(r+k)$, and $\gcd(m, 4sr) = 1$, so

$$m \mid r + k.$$

Since $k \geq r \geq 1$, the least admissible k is at least $\max(r, m-r) \geq m/2$, so $\sum_{k \equiv -r(m), k \leq X} 1/k \leq (3 + \log X)/m$. Summing over $m \mid 4sr^2 + 1$ gives the first bound; the second follows from $\sum_{s \leq X} 1/s \cdot \sum_{r \leq \sqrt{X}} 1/r \leq (1 + \mathcal{L})^2$ and $4sr^2 + 1 \leq T + 2$. $\square$

Remark 3.4. Without the congruence $m \mid r + k$ the same count gives $S \leq T^{1/2-\eta+o(1)}$ for $y = T^{1/2+\eta}$, and the transfer below then only yields $W(p) \gg \log p$. Numerically $S \approx 6.5, 13.6, 23.9, 38.5$ for $T = 10^3, \dots, 10^6$ (with $y = \sqrt{T}$), i.e. S is polylogarithmic in practice. These values illustrate S only: the parameter $y = \sqrt{T}$ does *not* instantiate Theorem 6.1. At $y = \sqrt{T}$ one has $\max g \approx 0.08\text{--}0.33 > 1/16$, and the twist condition of §6 can fail (numerically it does at $T = 10^4$). The bound $g \leq 1/16$ of Lemma 3.2, and hence Theorem 6.1, is asymptotic: with the y above, $y < T$ (so that $\mathcal{U} \neq \emptyset$) only when $\log T > e^6 \approx 403$.

4. THE ANALYTIC INPUT

Theorem 4.1 (Thorner–Zaman [15, Cor. 1.4 and Rem. 1.5]; cited). *There are absolute effectively computable constants $c_4 > 0$, C_0 such that for coprime*

a, q with $q \geq 2$ and $x \geq q^{12}$,

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \log p = \frac{\lambda x}{\varphi(q)} (1 + \varepsilon),$$

$$|\varepsilon| \leq C_0 \left(e^{-c_4 \log x / \log q} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}} \right),$$

where $\lambda = 1 - \chi_1(a)x^{\beta_1-1}/\beta_1$ if the exceptional zero β_1 for the modulus q exists and $\lambda = 1$ otherwise. Here, following McCurley [9] as quoted in [15, p. 1], $\prod_{\chi \bmod q} L(s, \chi)$ has no zero in $\Re s \geq 1 - 1/(13 \log(q(|\Im s| + 3)))$ apart from at most one real zero β_1 , which is simple and belongs to a unique real character χ_1 ; also $0 < \lambda < 2$.

We used the arXiv version; we did not compare the numbering with the published version. We also use the quoted McCurley statement for moduli other than those of Theorem 4.1; alternatively the classical Landau–Page theorem [3, Ch. 14] with a constant $c \leq 1/13$ in place of $1/13$ works verbatim.

Lemma 4.2 (One severe character per family; proved modulo the McCurley statement). *Let $\mathcal{Q}$ be a finite set of moduli, all $\leq Z$. Call (χ^*, β) severe if χ^* is a real primitive character whose conductor q^* divides some $q \in \mathcal{Q}$, $L(\beta, \chi^*) = 0$ and $\beta \geq 1 - 1/(13 \log(3Z^2))$. Then there is at most one severe pair. If (χ^*, β^*) is severe and $q^* \mid q \in \mathcal{Q}$, then $\beta_1 = \beta^*$ and χ_1 is induced by χ^* for the modulus q . If $q \in \mathcal{Q}$ and no severe pair has $q^* \mid q$, then any exceptional β_1 of q satisfies $x^{\beta_1-1}/\beta_1 \leq 2 \exp(-\log x/(13 \log 3Z^2))$.*

Proof. If $(\chi_a, \beta_a), (\chi_b, \beta_b)$ are severe with conductors dividing $q_a, q_b \in \mathcal{Q}$, put $q' = \text{lcm}(q_a, q_b) \leq Z^2$. Both characters induce characters mod q' with the same zeros in $\Re s > 0$, and both β lie in McCurley's region for q' ; uniqueness there gives $\beta_a = \beta_b$ and $\chi_a = \chi_b$. The second claim holds because β^* lies in McCurley's region for $q \leq Z$. For the third, an exceptional zero of q in $[1 - 1/(13 \log 3Z^2), 1)$ would be severe; also $\beta_1 \geq 1 - 1/(13 \log 6) > 1/2$. $\square$

5. A TRANSFER THEOREM

Theorem 5.1 (Proved modulo Theorem 4.1). *Let $T \geq 1$, $Q \geq 2$, and let $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ be a finite real combination with $\gcd(d_i, Q) = \gcd(b_i, d_i) = 1$ and $B(n) \leq \mathbf{1}[W(n) > T]$ for all $n \equiv 1 \pmod{Q}$. Put $Z = Q \max d_i$, $\mu = \sum c_i / \varphi(d_i)$, $M_1 = \sum |c_i| / \varphi(d_i)$, and assume $\mu > 0$ and the twist condition: for every real primitive character ψ of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some i ,*

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i) / \varphi(d_i).$$

Let $K = 1 + \log(M_1/\mu)$. There is an absolute C_1 such that if $\log x \geq C_1 K \max(\log Z, K)$, then some prime $p \leq x$, $p \equiv 1 \pmod{Q}$, has $W(p) > T$.

Proof. Let $q_i = Qd_i$ and let a_i be the class $\equiv 1 \pmod{Q}$, $\equiv b_i \pmod{d_i}$; it is a unit class. Then $\sum_{p \leq x, p \equiv 1 \pmod{Q}} \log p B(p) = \sum_i c_i \theta(x; q_i, a_i)$. Since $M_1 \geq \mu$ we have $K \geq 1$, so taking $C_1 \geq 12$ gives $\log x \geq 12 \log Z$, i.e. $x \geq Z^{12} \geq q_i^{12}$, and Theorem 4.1 applies to every term. Put $E_1 = e^{-c_4 \log x / \log Z} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}}$ and $E_2 = e^{-\log x / (13 \log 3 Z^2)}$. Since $\gcd(Q, d_i) = 1$ we have $\varphi(q_i) = \varphi(Q)\varphi(d_i)$, and $q_i \leq Z$, so

$$\sum_i c_i \theta(x; q_i, a_i) = \frac{x}{\varphi(Q)} \sum_i \frac{c_i \lambda_i (1 + \varepsilon_i)}{\varphi(d_i)}, \quad |\varepsilon_i| \leq C_0 E_1,$$

with λ_i the factor of Theorem 4.1 for (q_i, a_i) . Apply Lemma 4.2 to $\mathcal{Q} = \{q_i\}$; a severe β satisfies $\beta \geq 1 - 1/(13 \log 6) > 1/2$. Write the conductor of a severe χ^* as $q^* = q_Q^* q^{*'} with q_Q^* supported on the primes of Q and $\gcd(q^{*'}, Q) = 1$, and $\chi^* = \chi_Q^* \chi^{*'}$.$

Case A: a severe pair exists with $q^ \mid Q$.* Then $q^* \mid q_i$ for every i . By the second claim of Lemma 4.2, every λ_i is formed with the same pair (χ^*, β^*) , so $\lambda_i = 1 - \chi^*(a_i) x^{\beta^*-1} / \beta^* = 1 - x^{\beta^*-1} / \beta^* =: \lambda_*$, because $a_i \equiv 1 \pmod{q^*}$. Moreover $\lambda_* > 0$: since $\log x \geq 12 \log Z > 2$ and $1/2 < \beta^* < 1$, $\log \beta^* \geq 1 - 1/\beta^* > 2(\beta^* - 1) > (\beta^* - 1) \log x$, i.e. $x^{\beta^*-1} / \beta^* < 1$. So $\sum c_i \theta \geq (x \lambda_* / \varphi(Q))(\mu - C_0 E_1 M_1)$.

Case B: otherwise. If a severe χ^* exists, then q^* divides some Qd_j with $\gcd(d_j, Q) = 1$, so $q_Q^* \mid Q$. Hence $q^* \mid q_i$ iff $q^{*' \mid d_i$, and in Case B we have $q^{*' > 1$. For such i , $\chi_1 = \chi^*$ on units mod q_i (Lemma 4.2), and, as $a_i \equiv 1 \pmod{q_Q^*}$ and $a_i \equiv b_i \pmod{q^{*'}}$, $\chi^*(a_i) = \chi_Q^*(1) \chi^{*'}(b_i) = \psi(b_i)$ with $\psi = \chi^{*'}$. So $\lambda_i = 1 - \psi(b_i) x^{\beta^*-1} / \beta^*$. This ψ is real primitive (the coprime-conductor factors of a real primitive character are real primitive), of conductor $q^{*' > 1$ coprime to Q and dividing some d_i , so the twist condition applies to it. For all other i (and for all i if there is no severe pair), $|\lambda_i - 1| \leq x^{\beta_1-1} / \beta_1 \leq 2E_2$ by the third claim of Lemma 4.2. Since $\lambda_i < 2$ and $x^{\beta^*-1} / \beta^* \leq 1$, splitting $\lambda_i = 1 + (\lambda_i - 1)$ gives

$$\begin{aligned} \sum c_i \theta &\geq \frac{x}{\varphi(Q)} (\mu - 2|\mu_\psi| - M_1(2C_0 E_1 + 2E_2)) \\ &\geq \frac{x}{\varphi(Q)} (\mu/2 - M_1(2C_0 E_1 + 2E_2)), \end{aligned}$$

using $|\mu_\psi| \leq \mu/4$. Both brackets are positive once $E_1, E_2 \leq \mu/(8(C_0 + 1)M_1) = e^{1-K}/(8(C_0 + 1))$. This holds if C_1 is large: $\log x / \log Z \geq C_1 K$, $(\log x)^{3/5} / (\log \log x)^{1/5} \geq (\log x)^{1/2} \geq C_1^{1/2} K$ and $\log x / (13 \log 3 Z^2) \geq C_1 K / 52$ (as $Z \geq 2$), so each of the three exponentials is at most $e^{-c C_1^{1/2} K}$ with $c > 0$ absolute. Hence $\sum_p \log p \mathbf{1}[W(p) > T] \geq \sum_p \log p B(p) > 0$. $\square$

No lower bound for $1 - \beta^*$ is used: in Case A the exceptional factor λ_* multiplies main term and error alike. A prime number theorem valid only for $q \leq \exp(c\sqrt{\log x})$ would force $\log x \gg (\log Z)^2 \geq T$; the Linnik-range uniformity is essential.

6. WARM-UP: EXPONENT TWO BY A PRIME-LOCAL DESIGN

Theorem 6.1 (Proved modulo Theorem 4.1; effective). *For infinitely many hard primes p , $W(p) \geq (\log p)^2 \exp(-C \log \log p / \log \log \log p)$. More precisely, for every large T there is a prime $p \equiv 1 \pmod{Q}$ (Q as in §3) with $W(p) > T$ and $\log p \leq T^{1/2} \exp(O(\log T / \log \log T))$.*

Theorem 1.1 supersedes this; we include it because its minorant is a plain Brun truncation and it isolates the arithmetic input (Lemma 3.3) that the main proof reuses.

Proof. Let J be the least even integer $\geq 22S + 10$ and

$$B(n) = \sum_{P \subseteq \mathcal{U}, |P| \leq J-1} (-1)^{|P|} \prod_{\ell \in P} \mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{j=0}^{J-1} (-1)^j \binom{N(n)}{j},$$

$$N(n) = \#\{\ell \in \mathcal{U} : n \bmod \ell \in F_\ell\}.$$

Pointwise bound. For $N \geq 1$, $\sum_{j \leq J-1} (-1)^j \binom{N}{j} = (-1)^{J-1} \binom{N-1}{J-1} \leq 0$; hence $B \leq \mathbf{1}[N = 0] \leq \mathbf{1}[W > T]$ on $n \equiv 1 \pmod{Q}$ by Lemma 3.1. Expanding $\mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{a \in F_\ell} \mathbf{1}[n \equiv a \pmod{\ell}]$ puts B in the form of Theorem 5.1, with squarefree moduli $d_P = \prod_{\ell \in P} \ell \leq T^J$ coprime to Q .

Mean and mass. With e_j the elementary symmetric functions of the $g(\ell)$, $\mu = \sum_{j < J} (-1)^j e_j$ and $M_1 = \sum_{j < J} e_j \leq e^S$. Let N be a sum of independent Bernoulli($g(\ell)$) variables; the identity above and $\binom{N-1}{J-1} \leq \binom{N}{J}$ give $|\mu - V| \leq e_J \leq (eS/J)^J \leq e^{-2J}$, where $V = \prod (1 - g(\ell)) \geq e^{-2S}$ (Lemma 3.2). So $\mu \geq 0.99V$ and $\log(M_1/\mu) \leq 3S + 1$.

Twist. If ψ is real primitive of conductor $f > 1$, coprime to Q , dividing some d_P , then $f = \prod_{\ell \in P_f} \ell$ with $r = |P_f| \geq 1$ and $\psi = (\cdot/f)$. With $h(\ell) = \sum_{a \in F_\ell} (a/\ell)/(\ell - 1)$, $|h| \leq g$,

$$\mu_\psi = (-1)^r \prod_{\ell \in P_f} h(\ell) \sum_{P' \subseteq \mathcal{U} \setminus P_f, |P'| \leq J-1-r} (-1)^{|P'|} \prod_{P'} g,$$

which vanishes if $r \geq J$. The inner sum is $V' + \varepsilon'$ with $V' = V / \prod_{P_f} (1 - g) \leq (16/15)^r V$ and $|\varepsilon'| \leq e_{J-r} \leq S^{J-r}/(J-r)!$, so, with $g \leq 1/16$, $|\mu_\psi| \leq 15^{-r} V + 16^{-r} S^{J-r}/(J-r)!$. The first term is $\leq V/15$. If $r \leq J/2$, the second is $\leq (eS/(J-r))^{J-r} \leq (2eS/J)^{J/2} \leq (e/11)^{J/2} \leq e^{-0.69J} \leq e^{-15S-6}$, as $\frac{1}{2} \log(11/e) > 0.69$ and $J \geq 22S + 10$. If $r > J/2$, it is $\leq 16^{-J/2} e^S \leq e^{-1.38(22S+10)+S} \leq e^{-15S-6}$. Since $e^{-15S} \leq e^{-2S} \leq V$, we get $|\mu_\psi| \leq (1/15 + e^{-6})V \leq 0.07V \leq \mu/4$.

Moduli. $\log Z \leq \log Q + J\mathcal{L} \leq 3y$ because $S = T^{o(1)}$ (Lemma 3.3). Theorem 5.1 gives a prime $p \equiv 1 \pmod{Q}$ with $W(p) > T$ and $\log p \leq C_1(3S+2) \cdot 3y \leq \sqrt{T} \exp((3+\log 2+o(1))\mathcal{L}/\log \mathcal{L})$. Since $840 \mid Q$ the prime is hard, and $p > Q > T$ gives infinitely many distinct primes. Inverting: if $\mathcal{L} \geq 3 \log \log p$ then already $W(p) > T \geq (\log p)^3$; otherwise $\mathcal{L}/\log \mathcal{L} \ll$

$\log \log p / \log \log \log p$ and $\mathcal{L} \geq 2 \log \log p - O(\log \log p / \log \log \log p)$. All inputs are effective. $\square$

7. SUPPORT-TRUNCATED INCLUSION-EXCLUSION

Setting. Let $\mathcal{P}$ be a finite set of primes and $X = (X_\ell)_{\ell \in \mathcal{P}}$ independent random variables. Let $\mathcal{E}$ be a finite family of events; each $E \in \mathcal{E}$ has a nonempty support $\text{supp } E \subseteq \mathcal{P}$ and depends only on $(X_\ell)_{\ell \in \text{supp } E}$. For an outcome x put

$$A(x) = \{E \in \mathcal{E} : E \text{ occurs}\}, \quad V(x) = \bigcup_{E \in A(x)} \text{supp } E, \quad N(x) = |V(x)|,$$

$$a_\ell(x) = \#\{E \in A(x) : \ell \in \text{supp } E\}.$$

For $L \geq 0$ and $F \subseteq \mathcal{E}$ with $\text{supp } F = \bigcup_{E \in F} \text{supp } E$, define

$$B_L(x) = \sum_{F \subseteq A(x), |\text{supp } F| \leq L} (-1)^{|F|},$$

$$G_{L+1}(x) = e_{L+1}(a(x)) = \sum_{P \subseteq \mathcal{P}, |P|=L+1} \prod_{\ell \in P} a_\ell(x).$$

The truncation is by the number of *primes* involved, not by the number of events; this is what defeats the hub configurations of Proposition 11.6.

Lemma 7.1 (Proved). *Call $W \subseteq \mathcal{P}$ independent at x if no $E \in A(x)$ has $\text{supp } E \subseteq W$. If $A(x) = \emptyset$ then $B_L(x) = 1$. Otherwise*

$$B_L(x) = \sum_{\substack{W \subsetneq V(x) \\ |W| \leq L}} (-1)^{L-|W|} \binom{N-|W|-1}{L-|W|},$$

and in particular $B_L(x) = 0$ whenever $1 \leq N(x) \leq L$.

Proof. Fix x , write A, V, N . For $U \subseteq V$ the alternating sum over all $F \subseteq \{E \in A : \text{supp } E \subseteq U\}$ is $I(U) = \mathbf{1}[\text{no } E \in A \text{ has } \text{supp } E \subseteq U]$, so by Möbius inversion on the Boolean lattice $\sum_{F \subseteq A, \text{supp } F=U} (-1)^{|F|} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} I(W)$. Summing over $U \subseteq V$ with $|U| \leq L$,

$$B_L = \sum_{W \subseteq V, I(W)=1, |W| \leq L} \sum_{j=0}^{L-|W|} (-1)^j \binom{N-|W|}{j}.$$

The inner sum is $(-1)^{L-|W|} \binom{N-|W|-1}{L-|W|}$ if $|W| < N$ and 1 if $W = V$; and $I(V) = 1$ iff $A = \emptyset$. If $N \leq L$ every binomial $\binom{N-|W|-1}{L-|W|}$ vanishes. $\square$

Lemma 7.2 (Proved). *For every x and $L \geq 0$,*

$$B_L^*(x) := B_L(x) - 4^{L+1} G_{L+1}(x) \leq \mathbf{1}[A(x) = \emptyset],$$

$$|B_L(x) - \mathbf{1}[A(x) = \emptyset]| \leq 4^{L+1} G_{L+1}(x).$$

Proof. If $A = \emptyset$ then $a \equiv 0$ and $B_L = 1$. Let $A \neq \emptyset$. If $N \leq L$ then $B_L = 0$. If $N \geq L + 1$, Lemma 7.1 and Vandermonde give $|B_L| \leq \sum_{w \leq L} \binom{N}{w} \binom{N}{L-w} = \binom{2N}{L}$. For $f(N) = \binom{2N}{L} / \binom{N}{L+1}$ one computes

$$\frac{f(N+1)}{f(N)} = \frac{2(2N+1)(N-L)}{(2N+2-L)(2N+1-L)},$$

and with $t = N - L \geq 1$ the denominator exceeds the numerator by $L^2 + 3L + 4t + 2$; so $\binom{2N}{L} \leq f(L+1) \binom{N}{L+1} \leq 4^{L+1} \binom{N}{L+1}$. Finally $\binom{N}{L+1} = e_{L+1}(\mathbf{1}[\ell \in V]) \leq e_{L+1}(a)$. $\square$

For a finite combination $\Phi = \sum_i c_i \mathbf{1}_{\mathcal{C}_i}$ of indicators of cells $\mathcal{C}_i$ (a cell being a product of single values of finitely many coordinates) we write $M_1(\Phi) = \sum_i |c_i| P(\mathcal{C}_i)$, after merging equal cells; for congruence classes this is the M_1 of Theorem 5.1.

Lemma 7.3 (Proved). *Let $z \geq 16$ and $\Lambda_z = \log E \prod_{\ell \in \mathcal{P}} (1 + za_\ell)$. Then $E G_{L+1} \leq z^{-L-1} e^{\Lambda_z}$ and $E|B_L^* - \mathbf{1}[A = \emptyset]| \leq 2 \cdot 4^{L+1} z^{-L-1} e^{\Lambda_z}$; for $z = 16$ this is $2 \cdot 4^{-L-1} e^{\Lambda_{16}}$. If every event is a union of cells, then B_L is a combination of cells on supports of size $\leq L$ with $M_1(B_L) \leq E \prod_{\ell} (1 + 2a_\ell)$, and G_{L+1} is a nonnegative combination of cells on supports of size $\leq k(L+1)$, $k = \max |\text{supp } E|$, with $M_1(G_{L+1}) = E G_{L+1}$. If moreover no event contains a cell with a non-unit coordinate, then only unit cells have nonzero coefficient.*

Proof. Since all $a_\ell \geq 0$, we have $e_{L+1}(a) \leq z^{-L-1} \prod_{\ell} (1 + za_\ell)$ termwise. By Lemma 7.2,

$$E|B_L^* - \mathbf{1}[A = \emptyset]| \leq E|B_L - \mathbf{1}[A = \emptyset]| + 4^{L+1} E G_{L+1} \leq 2 \cdot 4^{L+1} E G_{L+1}.$$

Write $B_L = \sum_{F \subseteq \mathcal{E}, |\text{supp } F| \leq L} (-1)^{|F|} \prod_{E \in F} \mathbf{1}_E$ and group by $U = \text{supp } F$. On a cell c of the coordinates in U , let $A_U(c)$ be the set of events with support in U that contain c , and $I_c(W) = \mathbf{1}[\text{no } E \in A_U(c) \text{ has } \text{supp } E \subseteq W]$. The coefficient of c is

$$\kappa(U, c) = \sum_{F \subseteq A_U(c), \text{supp } F = U} (-1)^{|F|} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} I_c(W)$$

by the Möbius inversion of Lemma 7.1; so $|\kappa(U, c)| \leq 2^{|U|}$. If some $\ell \in U$ lies in the support of no event of $A_U(c)$, then $I_c(W) = I_c(W \cup \{\ell\})$ and the terms cancel in pairs, so $\kappa(U, c) = 0$; in particular $\kappa(U, c) = 0$ unless c is a unit cell, under the last hypothesis. For the remaining cells, every $\ell \in U$ is covered by an event occurring on c , so $\mathbf{1}_c \leq \prod_{\ell \in U} a_\ell$ on c . Hence

$$\begin{aligned} M_1(B_L) &\leq \sum_U 2^{|U|} P(\text{every } \ell \in U \text{ covered}) \\ &\leq \sum_U 2^{|U|} E \prod_{\ell \in U} a_\ell = E \prod_{\ell} (1 + 2a_\ell). \end{aligned}$$

Finally $G_{L+1} = \sum_{|P|=L+1} \prod_{\ell \in P} \sum_{E \ni \ell} \mathbf{1}_E$ expands into products of event indicators with coefficient $+1$, so its mass equals its mean. $\square$

8. A PSEUDOFORREST BOUND FOR THE EXPONENTIAL MOMENT

Graph-type systems. Each X_ℓ takes values in a finite set Ω_ℓ . The events are of two kinds. *Singles:* one set $S_\ell \subseteq \Omega_\ell$ per ℓ , with $g_\ell = P(X_\ell \in S_\ell)$ and $S_1 = \sum_\ell g_\ell$. *Edges:* a *vertex* is a pair $v = (\ell, V_v)$ with $V_v \subseteq \Omega_\ell$, distinct vertices at the same ℓ being disjoint, and $p(v) = P(X_\ell \in V_v)$; an edge is an unordered pair $\{u, w\}$ of vertices at distinct primes, occurring iff $X_{\ell_u} \in V_u$ and $X_{\ell_w} \in V_w$. The edge graph is simple: edges are distinct pairs of vertices (for congruences, distinct classes modulo $\ell\ell'$), repeated atoms being merged. Put

$$S_2 = \sum_{\text{edges}} p(u)p(w), \quad \deg(v) = \sum_{w: \{v, w\} \text{ edge}} p(w), \quad w_\ell = \sum_{\text{edges at } \ell} p(u)p(w).$$

Then $a_\ell = s_\ell + d_\ell$ with $s_\ell = \mathbf{1}[X_\ell \in S_\ell]$ and d_ℓ the number of occurring edges at ℓ .

Lemma 8.1 (Proved). *Let $z \geq 1$ and suppose $\deg(v) \leq \delta \leq e^{-3z-2}$ for every vertex. Then $\Lambda_z \leq zS_1 + 16e^{6z+2}S_2$.*

Proof. Use $1 + z(s + d) \leq (1 + zs)(1 + zd)$ and expand $\prod_\ell (1 + zd_\ell) = \sum_{(U, f)} z^{|U|} \mathbf{1}[f(\ell) \text{ occurs for all } \ell \in U]$, where f assigns to each $\ell \in U$ an edge at a vertex of ℓ . Let $F = f(U)$. If F occurs, its vertices lie at distinct primes, and each component K of F has $|E(K)| \leq |V(K)|$ (each edge of K is chosen by a prime whose vertex lies in K); so F is a pseudoforest. For fixed F , the weight of the pairs (U, f) with $f(U) = F$ is at most $\prod_{v \in V(F)} (1 + z \deg_F(v)) \leq e^{2z|V(F)|}$. For the singles, independence gives $E[\prod_{\ell \in U_s} s_\ell \mathbf{1}[F \text{ occurs}]] \leq \prod_{\ell \in U_s \setminus \pi(F)} g_\ell \cdot P(F \text{ occurs})$, $\pi(F)$ being the primes of $V(F)$; summing $z^{|U_s|}$ over U_s costs $(1 + z)^{|\pi(F)|} \prod_\ell (1 + zg_\ell)$. Hence

$$\begin{aligned} E \prod_\ell (1 + za_\ell) &\leq e^{zS_1} \sum_F e^{3z|V(F)|} P(F \text{ occurs}) \\ &\leq e^{zS_1} \exp \left(\sum_K e^{3z|V(K)|} P(K \text{ occurs}) \right), \end{aligned}$$

K running over connected pseudoforests; here we used that $P(F)$ is the product over components when these lie on disjoint primes and 0 otherwise. A connected pseudoforest on v vertices is a spanning tree T plus at most one of $\leq v^2/2$ further pairs, and $P(K) \leq P(T) \leq \prod_{w \in T} p(w)$. Encode a tree rooted at v_0 by its child sets $C_1, \dots, C_v$ in breadth-first order (children ordered by a fixed total order); every tree is recovered once for each choice of root, so summing over roots overcounts by the factor v . Summing over $C_i \subseteq N(v_i)$ with $|C_i| = c_i$ gives at most $\deg(v_i)^{c_i}/c_i!$, and for $i = 1$, $c_1 \geq 1$, at most $\deg(v_0)\delta^{c_1-1}/c_1!$; and $\sum_{c_1+\dots+c_v=v-1} \prod 1/c_i! = v^{v-1}/(v-1)! \leq e^v$. So the trees with $v \geq 2$ vertices contribute at most $\sum_{v_0} p(v_0) \deg(v_0) e^v \delta^{v-2} = 2S_2 e^v \delta^{v-2}$, and

$$\sum_K e^{3z|V(K)|} P(K) \leq 2S_2 e^{6z+2} \sum_{j \geq 0} \left(3 + 2j + \frac{j^2}{2}\right) (e^{3z+1}\delta)^j \leq 16e^{6z+2}S_2,$$

as $e^{3z+1}\delta \leq e^{-1}$ and $\sum_j(3+2j+j^2/2)e^{-j} < 7.6$. $\square$

The hypothesis is a bound on *vertex degrees*; configurations in which a complete bipartite graph of events fires (the hubs of Proposition 11.6) are not excluded. It is enforced as follows.

Lemma 8.2 (Proved). *For $\delta > 0$ let $H = \{v : \deg(v) > \delta\}$. Replace S_ℓ by $S_\ell^+ = S_\ell \cup \bigcup_{v \in H \text{ at } \ell} V_v$ and delete the edges with an endpoint in H . Then the absence of every new event implies the absence of every old event; every remaining vertex has degree $\leq \delta$; $g_\ell^+ \leq g_\ell + w_\ell/\delta$, $S_1^+ \leq S_1 + 2S_2/\delta$ and $S_2^+ \leq S_2$.*

Proof. An old edge at $v \in H$ cannot occur without the new single at ℓ_v . Degrees only decrease. By Markov, $\sum_{v \in H \text{ at } \ell} p(v) \leq \sum_{v \text{ at } \ell} p(v) \deg(v)/\delta = w_\ell/\delta$, and $\sum_\ell w_\ell = 2S_2$. $\square$

9. AN ABSTRACT CRITERION

Lemma 9.1 (Local lemma with conditioning; classical, proof included). *Let $\mathcal{A}$ be finitely many events in a product probability space, each depending on a finite set of coordinates; write $A' \sim A$ if they share a coordinate. Suppose $x_A \in [0, 1)$ satisfy $P(A) \leq x_A \prod_{A' \sim A} (1 - x_{A'})$. Then $P(\bigcap_{\mathcal{A}} \bar{A}) \geq \prod_{\mathcal{A}} (1 - x_A)$, and for every event B depending on a set of coordinates, $P(B \mid \bigcap_{\mathcal{A}} \bar{A}) \leq P(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$, where $\Gamma(B)$ is the set of $A \in \mathcal{A}$ sharing a coordinate with B .*

Proof. The first statement is [1, Lemma 5.1.1], whose proof shows $P(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \leq x_A$ for every $\mathcal{S} \subseteq \mathcal{A}$ with $A \notin \mathcal{S}$. For the second (cf. [6, Thm. 2.1]), B is independent of $\bigcap_{A \notin \Gamma(B)} \bar{A}$, so

$$P\left(B \mid \bigcap_{\mathcal{A}} \bar{A}\right) \leq \frac{P(B) P(\bigcap_{A \notin \Gamma(B)} \bar{A})}{P(\bigcap_{\mathcal{A}} \bar{A})} = \frac{P(B)}{P(\bigcap_{\Gamma(B)} \bar{A} \mid \bigcap_{A \notin \Gamma(B)} \bar{A})},$$

and the chain rule with the displayed claim bounds the denominator below by $\prod_{\Gamma(B)} (1 - x_A)$. $\square$

Congruence systems. Fix Q with $2 \mid Q$, a finite set $\mathcal{P}$ of primes coprime to Q , and exponents $e_\ell \geq 1$; put $X_\ell = n \bmod \ell^{e_\ell}$. The Haar measure is the uniform measure on $\prod_\ell (\mathbb{Z}/\ell^{e_\ell})^\times$; in it Theorem 5.1 computes μ . A single is a set S_ℓ of unit classes mod ℓ^{e_ℓ} , a vertex a unit class mod ℓ (so $p(v) = 1/(\ell - 1)$), an edge a unit class mod $\ell\ell'$. The pointwise statements of §7 hold for every integer n (non-unit residues trigger no event).

Theorem 9.2 (Proved). *Let $\delta = e^{-50}$ and consider a graph-type congruence system with*

- (G) $g_\ell \leq 1/32$ for all $\ell \in \mathcal{P}$;
- (W) $w_\ell \leq \delta/32$ for all $\ell \in \mathcal{P}$;
- (I) *for every integer $n \equiv 1 \pmod{Q}$ at which no single and no edge occurs, $W(n) > T$.*

Put $\Sigma = S_1 + S_2$. Then there is a minorant B as in Theorem 5.1 (moduli coprime to Q , unit classes, $B(n) \leq \mathbf{1}[W(n) > T]$ for $n \equiv 1 \pmod{Q}$) whose moduli are products of at most $C(\Sigma + 1)$ of the $\ell^{e\ell}$, with $\mu \geq \exp(-C(\Sigma + 1))$, $\log(M_1/\mu) \leq C(\Sigma + 1)$, and $|\mu_\psi| \leq \mu/4$ for every real primitive ψ as in Theorem 5.1. Here C is absolute.

Proof. Hubs. Apply Lemma 8.2 with δ : by (W), $g_\ell^+ \leq 1/16$, $S_1^+ \leq S_1 + 2S_2/\delta$, all degrees are $\leq \delta$, and (I) persists. Let $A(n)$ be the occurring events of the new system.

Local lemma. Apply Lemma 9.1 to all singles and edges with $x_E = 2P(E) \leq 1/8$. For a single at ℓ the neighbours are the edges at ℓ and $\prod(1-x) \geq 1 - \sum x \geq 1 - 2w_\ell \geq 1/2$; for an edge at ℓ, ℓ' the neighbours are the two singles and the edges at ℓ or ℓ' , and $\prod(1-x) \geq (7/8)^2(1-2w_\ell-2w_{\ell'}) \geq 1/2$. So the hypothesis of Lemma 9.1 holds, and since $1-x \geq e^{-1.1x}$ for $0 \leq x \leq 1/8$ and $\sum_E x_E = 2(S_1^+ + S_2^+) \leq 2(S_1^+ + S_2)$,

$$P(A = \emptyset) \geq \prod_E (1 - x_E) \geq e^{-\lambda}, \quad \lambda = 3(S_1^+ + S_2).$$

Minorant. With $\Lambda = \Lambda_{16} \leq 16S_1^+ + 16e^{98}S_2$ (Lemma 8.1 with $z = 16$, $\delta = e^{-50} = e^{-3z-2}$), let L be least with $4^{L+1} \geq 200e^{\Lambda+\lambda}$ and $B = B_L^*$ (Lemma 7.2), expanded into unit classes (Lemma 7.3; every event is a set of unit classes). Then $B \leq \mathbf{1}[A = \emptyset] \leq \mathbf{1}[W > T]$ on $n \equiv 1 \pmod{Q}$, by (I). By Lemma 7.3, $E|B - \mathbf{1}[A = \emptyset]| \leq 2 \cdot 4^{-L-1}e^\Lambda \leq e^{-\lambda}/100 \leq P(A = \emptyset)/100$, so $\mu = EB \geq 0.99e^{-\lambda}$; and $M_1 \leq M_1(B_L) + 4^{L+1}M_1(G_{L+1}) \leq e^\Lambda + 4^{L+1}16^{-L-1}e^\Lambda \leq 2e^\Lambda$, so $\log(M_1/\mu) \leq \Lambda + \lambda + 1$. Each term involves at most $2(L+1)$ primes, and Λ, λ, L are $O(\Sigma + 1)$.

Twist. Let ψ be real primitive of conductor $f > 1$, $\gcd(f, Q) = 1$, $f \mid d_i$ for some i . As $2 \mid Q$, f is odd, hence squarefree, and its primes lie in $\mathcal{P}$; fix one, ℓ_0 . If $f \nmid d_i$, some prime of f does not divide d_i and the character sum over that coordinate vanishes; hence $\mu_\psi = E[B\psi]$ (Haar) and

$$|\mu_\psi| \leq E|B - \mathbf{1}[A = \emptyset]| + |E[\mathbf{1}[A = \emptyset]\psi]|.$$

Let $\bar{A}'$ be the event that no event whose support avoids ℓ_0 occurs; it depends on $X_{-\ell_0} = (X_\ell)_{\ell \neq \ell_0}$ only. Then $\mathbf{1}[A = \emptyset] = \mathbf{1}_{\bar{A}'} \cdot \mathbf{1}[X_{\ell_0} \notin \text{Forb}]$, where $\text{Forb} = \text{Forb}(X_{-\ell_0})$ is the union of $S_{\ell_0}^+$ and of the vertices at ℓ_0 joined to an occurring vertex. Write $\psi = \chi_0(X_{\ell_0})\psi'(X_{-\ell_0})$ with $\chi_0 = (\cdot/\ell_0)$. Conditioning on $X_{-\ell_0}$ and using independence and $E\chi_0(X_{\ell_0}) = 0$,

$$|E_{X_{\ell_0}}[\mathbf{1}[X_{\ell_0} \notin \text{Forb}]\chi_0]| = |E_{X_{\ell_0}}[\mathbf{1}[X_{\ell_0} \in \text{Forb}]\chi_0]| \leq P_{X_{\ell_0}}(\text{Forb}),$$

so $|E[\mathbf{1}[A = \emptyset]\psi]| \leq P(\bar{A}' \cap \{X_{\ell_0} \in \text{Forb}\})$, which is at most

$$P(\bar{A}') \left(g_{\ell_0}^+ + \sum_{\{u,w\}, u \text{ at } \ell_0} p(u)P(w \mid \bar{A}') \right) \leq P(\bar{A}') \left(\frac{1}{16} + 1.22w_{\ell_0} \right) < 0.064 P(\bar{A}').$$

Here the single at ℓ_0 and X_{ℓ_0} are independent of $\bar{A}'$ and of X_{ℓ_w} , and Lemma 9.1, applied to the family defining $\bar{A}'$ and $B = \{X_{\ell_w} \in V_w\}$ (neighbours: the single and the edges at ℓ_w , with $\prod(1-x) \geq (7/8)(15/16) > 0.82$),

gives $P(w \mid \bar{A}') \leq 1.22p(w)$. As $P(A = \emptyset) = P(\bar{A}') - P(\bar{A}' \cap \{X_{\ell_0} \in \text{Forb}\}) \geq 0.936P(\bar{A}')$, we get $|\mu_\psi| < 0.08P(A = \emptyset) < \mu/4$. $\square$

10. PROOF OF THEOREM 1.1

For a set Π of primes and $M \leq T$, $M \equiv 3 \pmod{4}$, write $M = m_\Pi r_\Pi$ with m_Π the Π -part. An atom (M, D) , $D \mid A_M^2$, survives Π if $m_\Pi \mid 4D + 1$ and $r_\Pi > 1$; its Haar weight is $1/\varphi(r_\Pi)$.

Lemma 10.1 (Proved). *For every set Π of primes,*

$$S_{\text{tot}}(\Pi) = \sum_{\text{atoms surviving } \Pi} \frac{1}{\varphi(r_\Pi)} \leq \exp\left(O\left(\frac{\log T}{\log \log T}\right)\right).$$

Proof. Use $1/\varphi(r) \ll \log \log T \cdot m/M$ with $m = m_\Pi$ and follow the proof of Lemma 3.3: the involution preserves $m \mid 4D + 1$ because $m \mid M$; writing $D = sr'^2$, $A = sr'k$ with $r' \leq k \leq X$ gives $m \mid r' + k$; for fixed (s, r') each k determines M and hence m , and $m \mid 4sr'^2 + 1$. So the sum is $\ll \log \log T (3 + \mathcal{L}) \sum_{sr'^2 \leq X} \tau(4sr'^2 + 1)/(sr')$. $\square$

Proposition 10.2 (Proved). *For all large T , with $y = T^{1/3} \exp(2\mathcal{L}/\log \mathcal{L})$, there are Q with $840 \mid Q$ and $\log Q \leq 5y$, and a minorant B as in Theorem 5.1 with $\mu > 0$, the twist condition, and $\log \max d_i$, $\log(M_1/\mu)$ both at most $\exp(O(\mathcal{L}/\log \mathcal{L}))$. In particular $H_{\min}(\theta)$ (Hypothesis 11.4) holds for every $\theta \geq 1/3$.*

Proof. Let $\tau^* = \max_{n \leq T+2} \tau(n)$, so that $\tau^* \leq \exp((\log 2 + o(1))\mathcal{L}/\log \mathcal{L})$, and

$$y = T^{1/3} \exp(2\mathcal{L}/\log \mathcal{L}), \quad \Pi_0 = \{\ell \leq y\}.$$

For $\ell > y$ let $g_\ell^{(0)}$ be the Haar measure of the classes mod ℓ^{e_ℓ} (e_ℓ maximal with $\ell^{e_\ell} \leq T$) hit by atoms surviving Π_0 with $r_{\Pi_0} \in \{\ell, \ell^2\}$, and let $\mathcal{B} = \{\ell > y : g_\ell^{(0)} > 1/64\}$, so $|\mathcal{B}| \leq 64S_{\text{tot}}(\Pi_0)$. Put $\Pi = \Pi_0 \cup \mathcal{B}$, $Q = \text{lcm}(24, \ell^{e_\ell} : \ell \in \Pi)$, and $\mathcal{P} = \{y < \ell \leq T\} \setminus \mathcal{B}$. Every atom surviving Π has $r_\Pi \leq T$ with all prime factors in $\mathcal{P}$, hence $> T^{1/3}$, so $r_\Pi \in \{\ell, \ell^2, \ell\ell'\}$: it contributes the class $-4D \bmod r_\Pi$ to the single S_ℓ (lifted mod ℓ^{e_ℓ}) or the edge $-4D \bmod \ell\ell'$.

(I). Let $n \equiv 1 \pmod{Q}$ and $M \leq T$, $M \equiv 3 \pmod{4}$. Every $\ell^v \parallel M$ with $\ell \in \Pi$ has $\ell^v \leq T$, so $m_\Pi \mid Q$. If $r_\Pi = 1$, Lemma 2.2 applies. If $n \equiv -4D \pmod{M}$ with $D \mid A_M^2$, then $m_\Pi \mid 4D + 1$, and reducing mod r_Π shows that a single or edge occurs. Conclude with Lemma 2.1.

(W). An edge at ℓ comes from $M = m\ell\ell'$ with $m \leq T/(\ell\ell')$, and each M carries $\leq \tau^{*2}$ atoms; so $w_\ell \leq \sum_{\ell' > y} (T/(\ell\ell')) \tau^{*2} \cdot 4/(\ell\ell') \leq 8\tau^{*2}T/y^3 = \exp((2\log 2 - 6 + o(1))\mathcal{L}/\log \mathcal{L})$.

(G). An atom with $r_\Pi \in \{\ell, \ell^2\}$ either survives Π_0 with the same rough part (weight $\leq 1/64$ in total) or has $M = m\ell\ell'$ with $b \in \mathcal{B}$ ($b\ell^2, b^2\ell > y^3 > T$). The latter add at most $2|\mathcal{B}|\tau^{*2}T/y^3 = \exp((3\log 2 - 6 + o(1))\mathcal{L}/\log \mathcal{L})$. So $g_\ell \leq 1/32$.

Sizes. $\Sigma \leq S_{\text{tot}}(\Pi) = \exp(O(\mathcal{L}/\log \mathcal{L}))$ by Lemma 10.1, and $\log Q \leq (\pi(y) + |\mathcal{B}|)\mathcal{L} + \log 24 \leq 5y$.

Theorem 9.2 now gives B with $K = 1 + \log(M_1/\mu)$ and $\log \max d_i$ both at most $\exp(O(\mathcal{L}/\log \mathcal{L}))$. The conditions of Hypothesis 11.4 hold for every $\theta \geq 1/3$ and $\varepsilon > 0$, since $\exp(O(\mathcal{L}/\log \mathcal{L})) = T^{o(1)}$. $\square$

Proof of Theorem 1.1. Take Q, B, y from Proposition 10.2 and put $K = 1 + \log(M_1/\mu)$. As in Theorem 11.5, adjoin to Q a prime $\ell_0 \in (R, 2R]$, $R = \max(T, \max d_i)$; then $\log Z \leq 6y$ and Theorem 5.1 gives a prime $p \equiv 1 \pmod{Q\ell_0}$ with $W(p) > T$ and $\log p \leq C_1 K \max(\log Z, K) \leq T^{1/3} \exp(O(\mathcal{L}/\log \mathcal{L}))$. It is hard as $840 \mid Q$, and $p > \ell_0 > T$ gives infinitely many distinct primes. Inverting: if $\mathcal{L} \geq 4 \log \log p$ then $W(p) > T \geq (\log p)^4$; otherwise $\mathcal{L}/\log \mathcal{L} \ll \log \log p / \log \log \log p$, and $\mathcal{L} \geq 3 \log \log p - O(\log \log p / \log \log \log p)$. All inputs are effective. $\square$

The constants are astronomically lossy ($\delta = e^{-50}$ in Theorem 9.2); the statement is purely asymptotic. Exponent 3 uses $y \geq T^{1/3+o(1)}$ twice: for the crude per-prime bounds (W), (G), and to ensure that all events have at most two primes.

11. LIMITS OF THE METHOD

11.1. Complete certificates. A *complete certificate* for $W > T$ is a reduced class $c \pmod{Q}$ all of whose primes have $W(p) > T$. The class of one modulo $L^*(T) = \text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$ is one, with $\log L^*(T) = (2/3 + o(1))T$.

Proposition 11.1 (Proved). *Every complete certificate has $\log Q \geq (2/3 - o(1))T$.*

Proof. (i) If a prime $\ell \equiv 3 \pmod{4}$, $\ell \leq T$, does not divide Q , then c is compatible with the atom $-4 \in \mathcal{R}(\ell)$ ($D = 1$), and Dirichlet's theorem gives primes with $W \leq \ell$; so $\prod_{\ell \leq T, \ell \equiv 3(4)} \ell \mid Q$. (ii) Since $\mathcal{R}(3) = \{2\}$, $3 \mid Q$ and $c \equiv 1 \pmod{3}$. (iii) Let $\ell \equiv 1 \pmod{4}$, $\ell \leq T/3$, and suppose a prime $r \equiv 2 \pmod{3}$ divides $A = (3\ell + 1)/4$. The atom $M = 3\ell$, $D = r$ has class $-4r \equiv 1 \pmod{3}$; if $\ell \nmid Q$ it is compatible with c , a contradiction. (iv) The primes $\ell \equiv 1 \pmod{4}$ for which $(3\ell + 1)/4$ has no prime factor $\equiv 2 \pmod{3}$ avoid the classes $\ell \equiv -3^{-1} \pmod{r}$ for all primes $r \equiv 2 \pmod{3}$, $r \geq 5$; by the prime number theorem in progressions their logarithmic weight up to X is $o(X)$. Hence $\log Q \geq T/2 + T/6 - o(T)$. $\square$

Remark 11.2 (The previous record; proved modulo [2, Cor. 11]). Put $L^*(T) = \text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$. Every prime $p \equiv 1 \pmod{L^*(T)}$ has $W(p) > T$, by Lemmas 2.1 and 2.2. A prime $\ell \leq T/3$ divides some such M (namely ℓ or 3ℓ). A prime $\ell \in (T/3, T]$ divides one iff $\ell \equiv 3 \pmod{4}$. Higher prime powers contribute $O(\sqrt{T} \log T)$. Hence $\log L^*(T) = (2/3 + o(1))T$. Every prime factor ℓ of $L^*(T)$ has $\log \ell \leq \log T = o(\log L^*(T))$, so Chang's bound gives a prime $p \equiv 1 \pmod{L^*(T)}$ with $\log p \leq (12/5 + o(1)) \log L^*(T)$,

i.e. $W(p) > T \geq (5/8 - \varepsilon) \log p$. As $840 \mid L^*(T)$ and $p > L^*(T) > T$, this holds for infinitely many hard primes.

Proposition 11.1 bounds $\log Q$, not p : a class $c \neq 1$ may contain primes below Q . What it shows is that a complete certificate whose prime is supplied by a least-prime theorem, which certifies $\log p$ only through an upper bound $O(\log Q)$, certifies no more than $W(p) \gg \log p$; with Chang's exponent, no more than $5/8$ by this route.

11.2. Prime-local designs. Call a design *prime-local* if it imposes $n \equiv 1 \pmod{\ell^{e_\ell}}$ for ℓ in a set Π and every $M \leq T$, $M \equiv 3 \pmod{4}$, has at most one prime factor outside Π . Theorem 6.1 uses such a design.

Proposition 11.3 (Proved). *In a prime-local design, Π contains every odd prime $\leq \sqrt{T/3}$ with at most one exception. Hence every prime it produces has $\log p \geq (1/\sqrt{3} - o(1))\sqrt{T}$.*

Proof. For odd primes $\ell_1 < \ell_2 \leq \sqrt{T/3}$ outside Π , one of $\ell_1\ell_2, 3\ell_1\ell_2$ is $\equiv 3 \pmod{4}$ and $\leq T$. $\square$

11.3. The hypothesis $H_{\min}$ and the hub obstruction.

Hypothesis 11.4 ($H_{\min}(\theta)$; proved for $\theta \geq 1/3$ (Proposition 10.2), open for $\theta < 1/3$). For every $\varepsilon > 0$ and large T there are Q with $840 \mid Q$, $\log Q \leq T^{\theta+\varepsilon}$, and a minorant B as in Theorem 5.1 with $\log \max d_i \leq T^{\theta+\varepsilon}$, $\log(M_1/\mu) \leq T^\varepsilon$, $\mu > 0$, and the twist condition.

Theorem 11.5 (Proved modulo Theorem 4.1). *$H_{\min}(\theta)$ implies $W(p) > (\log p)^{1/\theta-\varepsilon}$ for infinitely many hard primes, for every $\varepsilon > 0$.*

Proof. Adjoin to Q a prime $\ell_0 \in (R, 2R]$, $R = \max(T, \max d_i)$; this keeps all hypotheses and forces $p > T$. Theorem 5.1 gives $\log p \leq T^{\theta+2\varepsilon+o(1)}$. $\square$

$H_{\min}$ is a purely combinatorial statement about congruences. The Proposition 10.2 (via Theorem 9.2) establishes it for every $\theta \geq 1/3$ (indeed with $T^{o(1)} = \exp(O(\log T / \log \log T))$). For $\theta < 1/2$ the residual atoms involve several free primes, and plain event-level Bonferroni truncation over the raw atom list fails:

Proposition 11.6 (Proved). *Fix $0 < \theta < 1/2$, put $y = T^\theta$, and impose the class of one modulo all prime powers $\leq T$ of primes $\leq y$. Take as events all atoms (M, c) , $c \in \mathcal{R}(M)$, compatible with this, and let $N(n)$ count those containing n . Then for large T and every even J with $(4.2\theta^2 + o(1))(\log T)^2 \leq J \leq y$, the Bonferroni minorant $B_J = \sum_{j < J} (-1)^j \binom{N}{j}$ has negative mean in the CRT measure on the class 1.*

Proof. The mean is $P(N=0) - E[(\binom{N-1}{j-1} \mathbf{1}_{N \geq 1}])$. Let $r = \lceil \sqrt{2J-1} \rceil \leq \sqrt{2y}+1$, and take r primes $\equiv 1 \pmod{4}$ and r primes $\equiv 3 \pmod{4}$ in $(y, 2y]$; there are enough of them. If $n \equiv -4$ modulo all of them, then for every cross pair

$M = \ell\ell' \leq 4y^2 \leq T$ we have $M \equiv 3 \pmod{4}$, and $D = 1$ gives the atom $-4 \pmod{M}$. So $N \geq r^2 \geq 2J - 1$, on a set of measure $\geq (2y)^{-2r}$. Hence

$$E\binom{N-1}{J-1} \geq (2y)^{-2r} \binom{2J-2}{J-1} \geq \exp((J-1)\log 4 - \log(2J-1) - 2r(\theta \log T + \log 2)),$$

which exceeds $1 > P(N = 0)$ once $J > (2\sqrt{2}\theta/\log 4)^2(\log T)^2(1 + o(1))$. $\square$

Proposition 11.6 is an obstruction only to truncation by the number of events. On the hub configuration of its proof, r^2 events fire on $2r$ primes; the error of the support truncation (Lemma 7.2) is at most $\binom{4r}{L} \leq 16^r$ there, against probability $(2y)^{-2r}$.

Below $\theta = 1/3$. The proof of Theorem 1.1 uses $y \geq T^{1/3+o(1)}$ for the crude per-prime bounds (W), (G) and to ensure that events involve at most two primes. For smaller θ one would need a per-prime mass bound of the kind discussed in §13 and a hypergraph version of Lemma 8.1; we have neither.

12. THE SLICE PARAMETER

For a prime p let $\mathcal{B}_p$ be the set of pairs (c, k) with $1 \leq k \leq \lfloor 2p/3 \rfloor$, $1 \leq c \leq \lfloor (2p+k)/(4k) \rfloor$, $(p, ck) = 1$; put $h = 4ck$, $N_{c,k} = p^2 + 4ck^2$, $s = \text{sf}(c)$, and let $\chi_s = (\Delta_s/\cdot)$, where $\Delta_s < 0$ is the discriminant of $\mathbb{Q}(\sqrt{-s})$; its conductor divides $4s \mid h$. $M_{c,k}(p)$ is the number of divisors of $N_{c,k}$ in the class $-p \pmod{h}$. The slice (c, k) is *positive* if $p(a+b) = k(4abc-1)$ has a solution in positive integers a, b . This happens iff $M_{c,k}(p) > 0$. Both directions rest on the identity, for $e = ha - p$, $f = hb - p$,

$$(ha-p)(hb-p) - N = h(4abck - p(a+b)) - 4ck^2 = h(k(4abc-1) - p(a+b)),$$

so $ef = N$ iff $k(4abc-1) = p(a+b)$. If $e \equiv -p \pmod{h}$ is a positive divisor of N , then $f = N/e \equiv p^2(-p)^{-1} \equiv -p \pmod{h}$ (as $\gcd(p, h) = 1$), so $a = (p+e)/h$, $b = (p+f)/h$ are positive integers with $ef = N$. Conversely, if $k(4abc-1) = p(a+b)$ then $pa, pb < 4abck$, so $e = ha - p > 0$ and $f = hb - p > 0$; and $ef = N$, $e \equiv -p \pmod{h}$.

Lemma 12.1 (Genus obstruction; proved). *If $p \equiv 1 \pmod{24}$, $(c, k) \in \mathcal{B}_p$ and $\chi_s(p) = 1$, then $M_{c,k}(p) = 0$.*

Proof. Write $c = st^2$. Every prime $q \mid N$ is odd, prime to stk , and $(p(2tk)^{-1})^2 \equiv -s \pmod{q}$, so $\chi_s(q) = 1$; hence $\chi_s(e) = 1$ for every divisor e of N . But $\chi_s(-p) = \chi_s(-1)\chi_s(p) = -1$, and χ_s is defined modulo h . $\square$

Lemma 12.2 (A sufficient prime factor; proved). *If a prime q divides $N_{c,k}$ and $q \equiv -p \pmod{h}$, then $M_{c,k}(p) \geq 1$.*

Proof. q itself is a divisor in the class $-p$. $\square$

Lemma 12.3 (Proved). *Let $p \equiv 1 \pmod{24}$ and suppose $(\ell/p) = 1$ for every prime $5 \leq \ell \leq B$. Then $ck_{\min}(p) > B$. In particular $ck_{\min}(p) \geq n_p$, the least quadratic non-residue modulo p .*

Proof. For $ck \leq B$, $s \leq B$ and $\chi_s(p) = (-s/p) = (-1/p) \prod_{q|s} (q/p) = 1$, since $(-1/p) = (2/p) = (3/p) = 1$. Apply Lemma 12.1. $\square$

Proposition 12.4 (Proved). *Let $24 \mid L$ and let $a \bmod L$ be a reduced class with $a \equiv 1 \pmod{24}$ such that every sufficiently large prime $p \equiv a \pmod{L}$ has $ck_{\min}(p) > T$. Then every prime $5 \leq \ell \leq T$ divides L and $(a/\ell) = 1$.*

Proof. Suppose $\ell \nmid L$ or $(a/\ell) = -1$. Choose $c_0 \bmod 4\ell$ with $c_0 \equiv a \pmod{\gcd(L, 4\ell)}$, $c_0 \equiv 1 \pmod{4}$, $(c_0/\ell) = -1$; for $n \equiv 1 \pmod{4}$, $\chi_\ell(n) = (n/\ell)$, so $\chi_\ell(c_0) = -1$. Take a prime $q \equiv -c_0 \pmod{4\ell}$, $q \nmid L$. Then $\chi_\ell(q) = \chi_\ell(-1)\chi_\ell(c_0) = 1$, so $-4\ell \equiv \rho^2 \pmod{q}$ for some $\rho \not\equiv 0$. The conditions $p \equiv a \pmod{L}$, $p \equiv c_0 \pmod{4\ell}$, $p \equiv \rho \pmod{q}$ are compatible and define a reduced class, which contains infinitely many primes. For these, $q \mid p^2 + 4\ell$ and $q \equiv -p \pmod{4\ell}$, so $M_{\ell,1}(p) \geq 1$ (Lemma 12.2), and $(\ell, 1) \in \mathcal{B}_p$ for large p , whence $ck_{\min}(p) \leq \ell \leq T$. $\square$

Corollary 12.5 (Proved). *Let $B(n)$ be a finite combination of residue classes with $B(p) \leq 1[ck_{\min}(p) > T]$ for all large primes p in a reduced class $\equiv 1 \pmod{24}$. Then every large prime p in that class with $B(p) > 0$ has $n_p > T$. Together with Lemma 12.3: the depth certifiable by congruences is exactly n_p .*

Proof. B is periodic modulo some L , which we take divisible by the modulus of the given class (hence by 24). If $B(p) > 0$ then B is positive on the whole class $p \bmod L$, which lies in the given class. So every large prime in the class $p \bmod L$ has $ck_{\min} > T$, and Proposition 12.4 gives $(p/\ell) = 1$, i.e. $(\ell/p) = 1$ by reciprocity ($p \equiv 1 \pmod{4}$), for $5 \leq \ell \leq T$; also $(2/p) = (3/p) = 1$ as $p \equiv 1 \pmod{24}$. Hence $n_p > T$. $\square$

Theorem 12.6 (Proved modulo [8, Prop. 5.1]). $ck_{\min}(p) \gg \log p \cdot \log \log \log p$ for infinitely many hard primes p .

Proof. Lau and Wu [8, Prop. 5.1] (following Graham–Ringrose [5]; the numbering is that of the authors’ preprint, and the published numbering was not compared) show the following. They phrase it via $\chi_p(q) = 1$ with $\chi_p = (p/\cdot)$, which for $p \equiv 1 \pmod{4}$ equals (q/p) by reciprocity, including $q = 2$. The statement: for small fixed $\delta > 0$ and $y(x) = \delta \log x \log_3 x$ there are $x_n \rightarrow \infty$ and primes $p \in (x_n^{1/2}, x_n \log x_n]$ with $p \equiv 1 \pmod{4}$ and $(q/p) = 1$ for all primes $q \leq y(x_n)$. Such p are $\equiv 1 \pmod{24}$ and, since $(5/p) = (7/p) = 1$, lie in a Mordell class. Lemma 12.3 gives $ck_{\min}(p) > y(x_n) \geq (\delta/2) \log p \log_3 p$. Effectivity is not claimed. $\square$

Under GRH, Montgomery’s $n_p = \Omega(\log p \log \log p)$ would give the corresponding bound, and Ankeny’s $n_p \ll (\log p)^2$ caps what congruences can certify; the random model predicts $\max_{p \leq x} n_p \asymp \log x \log \log x$ (statements as quoted in [8]).

Corollary 12.7 (Proved modulo Theorem 4.1). *For infinitely many hard primes p , jointly $W(p) \geq (\log p)^{3-o(1)}$ and $ck_{\min}(p) \geq (\log p)^{1-o(1)}$.*

Proof. The primes p of Theorem 1.1 satisfy $p \equiv 1 \pmod{Q}$, where $24 \mid Q$ and every prime $\ell \leq y = T^{1/3} \exp(2\mathcal{L}/\log \mathcal{L})$ divides Q (Proposition 10.2). So $p \equiv 1 \pmod{24}$ and $(p/\ell) = 1$, hence $(\ell/p) = 1$ by reciprocity ($p \equiv 1 \pmod{4}$), for every odd $\ell \leq y$. Lemma 12.3 gives $ck_{\min}(p) > y$, and $\log p \leq T^{1/3} \exp(O(\mathcal{L}/\log \mathcal{L}))$ gives $y \geq (\log p)^{1-o(1)}$. $\square$

13. THE HAAR SIDE

Impose the class of one at all primes $\leq z$ ($z \geq 3$), i.e. $n \equiv 1 \pmod{Q_z}$ with Q_z defined like Q . For $M \leq T$ write $M = mr$ with m the z -smooth and r the z -rough part. If $r > 1$, the surviving atoms of M are the $D \mid A_M^2$ with $m \mid 4D + 1$, each giving the event $n \equiv -4D \pmod{r}$ of conditional Haar probability $1/\varphi(r)$.

Lemma 13.1 (Proved). *If $M \equiv 3 \pmod{4}$, $D \mid A_M^2$ and $M = mr$ with $m \mid 4D + 1$, then $m \leq r^2 + 1$.*

Proof. By the involution of Lemma 3.3 assume $D \leq A$, $D = sr'^2$, $A = sr'k$, $k \geq r'$. As there, $m \mid r' + k \leq 2k$, so $r = (4sr'k - 1)/m \geq 2sr' - 1/(2k)$, i.e. $r \geq 2sr'$, and $m \leq 4sr'^2 + 1 \leq r^2 + 1$. $\square$

Lemma 13.2 (Proved). *For every $3 \leq z \leq T$, the total mass $S_{\text{tot}} = \sum 1/\varphi(r)$ over all surviving atoms is $\exp(O(\log T/\log \log T))$.*

Proof. Use $1/\varphi(r) \ll \log \log T \cdot m/M$ and repeat the proof of Lemma 3.3 with ℓ replaced by r ; every surviving atom has $m \mid \gcd(4D + 1, M)$, hence $m \mid r' + k$. $\square$

(Using [4, Prop. 1.4] on dyadic blocks one gets $S_{\text{tot}} \ll (\log T)^4 \log \log T$; this is not needed below.)

Proof of Theorem 1.3. Fix $\varepsilon > 0$, put $y = T^{1/3+\varepsilon}$ and $z = y$. Every rough part is ℓ , ℓ^2 or $\ell_1 \ell_2$ (as $y^3 > T$). Use the independent coordinates $X_\ell = n \pmod{\ell^{e_\ell}}$. Single atoms form forbidden sets G_ℓ of measure g_ℓ , with $1 \notin G_\ell$ (Lemma 2.2) and $\sum g_\ell \leq S_{\text{tot}}$.

Bad primes. $B = \{\ell : g_\ell > 1/4\}$ has $|B| \leq 4S_{\text{tot}} = T^{o(1)}$. Impose also $n \equiv 1 \pmod{\ell^{e_\ell}}$ for $\ell \in B$ (cost $\leq |B| \log T$). Pairs inside B die by Lemma 2.2; a pair atom $(\ell b, a)$ with $b \in B$ becomes the residue $a \pmod{\ell}$ at ℓ if $a \equiv 1 \pmod{b}$ and dies otherwise. The enlarged sets satisfy $g'_\ell \leq 1/4 + |B| \tau_{\max}^2 T/(\ell y(\ell - 1)) \leq 1/2$, with $\tau_{\max}^2 = \max_{A \leq T} \tau(A^2) = T^{o(1)}$, and $\sum (g'_\ell - g_\ell) \leq |B| \tau_{\max}^2 T/y^2 = T^{1/3-2\varepsilon+o(1)}$.

Good pairs. Under the product measure μ' uniform on the allowed residues, a good pair event has $\mu'(E) \leq 4/((\ell_1 - 1)(\ell_2 - 1))$, and the per-prime weight is $w'_\ell \leq 16\tau_{\max}^2 T/(\ell^2 y) \leq T^{-3\varepsilon+o(1)}$. The asymmetric local lemma [1, Lemma 5.1.1] applies to μ' . Here μ' is the Haar measure conditioned on a product set, hence a product measure, and a pair event is mutually independent of all events sharing no prime with it. With $x_E = 2\mu'(E)$ we have

$\sum_{E' \sim E} x_{E'} \leq 2(w'_{\ell_1} + w'_{\ell_2}) = T^{-3\varepsilon+o(1)}$, so $\prod_{E' \sim E} (1 - x_{E'}) \geq 1/2$ for large T . The lemma gives $\mu'(\text{no good pair event}) \geq \prod (1 - x_E) \geq \exp(-16S_{\text{tot}})$.

Collecting. With $Q' = Q_y \prod_{b \in B} b^{e_b}$,

$$\begin{aligned} \delta^*(T) &\geq \frac{8}{\varphi(Q')} \prod (1 - g'_\ell) e^{-16S_{\text{tot}}} \\ &\geq \exp(-\pi(y) \log T - |B| \log T - 18S_{\text{tot}} - T^{1/3-2\varepsilon+o(1)}) \\ &= \exp(-T^{1/3+\varepsilon+o(1)}). \end{aligned}$$

□

The exponent $1/3$ reflects the crude bound $N(M) \leq \tau(A^2)$ for multi-prime atoms at a fixed prime.

Let $w_\ell(T, z) = \sum 1/\varphi(r_E)$ over the distinct surviving events E whose rough part is divisible by ℓ .

Lemma 13.3 (Proved). *Let $3 \leq z \leq T$. If $w_\ell(T, z) \leq \log z / (8 \log T)$ for all primes $z < \ell \leq T$, then $\delta^*(T) \geq (8/\varphi(Q_z)) \exp(-4S_{\text{tot}})$, so $\log(1/\delta^*(T)) \leq \pi(z) \log T + 4S_{\text{tot}} + O(1)$, with S_{tot} as in Lemma 13.2.*

Proof. Apply the local lemma [1, Lemma 5.1.1] on the coordinates X_ℓ , $\ell > z$ (uniform on units), to all distinct surviving events, with $x_E = 2/\varphi(r_E) \leq 1/2$ and $E \sim E'$ when the rough parts share a prime. An event has at most $\log T / \log z$ primes, so $\prod_{E' \sim E} (1 - x_{E'}) \geq \exp(-4 \sum_{\ell | r_E} w_\ell) \geq e^{-1/2} \geq 1/2$, and the lemma gives probability $\geq \prod (1 - x_E) \geq \exp(-4S_{\text{tot}})$ of no event. The class of one modulo Q_z has relative measure $\varphi(24)/\varphi(Q_z)$, and $\log \varphi(Q_z) \leq \pi(z) \log T$. □

Remark 13.4 (Conditional; the implication is proved modulo [4, Prop. 1.4]). Suppose the hypothesis of Lemma 13.3 holds with $z = (\log T)^C$ for all large T . Then $\log(1/\delta^*(T)) \ll (\log T)^{\max(C+1, 4)+o(1)}$, since $S_{\text{tot}} \ll (\log T)^4 \log \log T$ by [4, Prop. 1.4]. Without [4], Lemma 13.2 gives only $\delta^*(T) \geq \exp(-T^{o(1)})$ by this argument. We do not know how to prove the per-prime hypothesis.

For $y = T^{1/3+o(1)}$ the Haar bound of Theorem 1.3 has a prime-side counterpart: Theorem 1.1, through the pointwise minorant of Theorem 9.2. The local lemma by itself does not provide a pointwise minorant.

Single-prime atoms and the Type I variety. Following [4, §2], let Σ_I^n be the set of sextuples (a, b, c, d, e, f) satisfying their identities (2.1)–(2.9), among them

$$4abd = ne + 1, \quad ce = a + b, \quad 4acd = n + f, \quad ef = 4a^2d + 1, \quad bf = na + c,$$

and put $F_I(n) = \#\{(a, b, c, d, e, f) \in \mathbb{N}^6 \cap \Sigma_I^n : a \leq b\}$.

Lemma 13.5 (Proved). *Let $r > 1$ be odd. The atoms (M, D) with $M = mr \equiv 3 \pmod{4}$, $D \mid A_M^2$, $m \mid 4D + 1$ and $D \leq A_M$ (any $m \geq 1$) correspond*

bijectively to the points of $\mathbb{N}^6 \cap \Sigma_I^r$ with $a \leq b$ and d squarefree, via $D = sr'^2$, $A_M = sr'k$ (s squarefree) and

$$(a, b, c, d, e, f) = (r', k, (r' + k)/m, s, m, (4D + 1)/m).$$

The class of the atom is $-4D \equiv -a/b \pmod{r}$. Hence the number of distinct classes modulo r carried by atoms with rough part r , under any class-of-one quarantine, is at most $2F_I(r)$.

Proof. By the proof of Lemma 3.3, $m \mid r' + k$, so $c \in \mathbb{N}$. Then $4abd = 4A_M = mr + 1$, $ce = a + b$ and $ef = 4sr'^2 + 1$; by direct computation these three identities (with nonzero entries) imply the others, e.g. $e \cdot 4acd = 4a^2d + 4abd = (ef - 1) + (ne + 1)$ and $e \cdot bf = a(ne + 1) + b = e(an + c)$. Conversely a point with $a \leq b$ and d squarefree gives $D = da^2 \leq A = dab$, $M = ne = 4A - 1$, $m = e \mid 4D + 1$. The triple (s, r', k) determines D and A , hence M ; and $4A \equiv 1 \pmod{r}$ gives $-4D \equiv -r'/k$. Partners A^2/D at most double the count. $\square$

So the single-prime forbidden sets are governed by the Type I variety Σ_I^r . Modulo the proof of [4, Prop. 1.7] (its Lemma 2.8 and §3), which bounds the relevant points, $F_I(n) \leq n^{3/5+O(1/\log \log n)}$; ET remark that $3/5$ appears to be the limit of divisor-bound arguments. The following elementary bound is not used below.

Proposition 13.6 (Proved). *For a prime ℓ , the points of $\mathbb{N}^6 \cap \Sigma_I^\ell$ with $a = 1$ number $\ll \sqrt{\ell} \log \ell$.*

Proof. A point is determined by (a, c, f) , via $4acd = \ell + f$, $ef = 4a^2d + 1$, $b = ce - a$. First $\ell \nmid c$: otherwise ℓ divides $y = acd$, $z = bcd$ and $x = abdl$ in $4/\ell = 1/x + 1/y + 1/z$, giving $4 \leq 3$. So $g = \gcd(c, f)$, which divides $\ell a + c$ (as $f \mid \ell a + c$ by $bf = \ell a + c$) and $\ell + f$ (as $c \mid \ell + f$), divides ℓ and hence equals 1. Then $f \mid \ell a + c$ and $c \mid a(\ell + f)$ give $cf \mid a(\ell + f) + c$, so $(c - a)(f - 1) \leq a(\ell + 1)$. With $a = 1$: either $f \leq 1 + \sqrt{\ell + 1}$, and then $c \mid \ell + f$; or $c \leq 1 + \sqrt{\ell + 1}$, and then $f \mid \ell + c$. So the count is at most $2 \sum_{j \leq 2 + \sqrt{\ell}} \tau(\ell + j)$, and $\tau(n) \leq 2\#\{\delta \mid n : \delta \leq \sqrt{n}\}$ gives $\sum_{j \leq 2 + \sqrt{\ell}} \tau(\ell + j) \leq 2 \sum_{\delta \leq \sqrt{2\ell}} (3 + \sqrt{\ell})/\delta + O(\sqrt{\ell}) \ll \sqrt{\ell} \log \ell$. $\square$

Theorem 13.7 (Proved implication). *Suppose $F_I(n) \leq n^{\eta+o(1)}$ for all n , for some $0 \leq \eta \leq 1$, and put $\kappa = \eta/(1 + \eta)$. Then $w_\ell(T, z) \leq T^{\kappa+o(1)}/\ell$ for all z and all primes $\ell \leq T$, and $\log(1/\delta^*(T)) \leq T^{\kappa+o(1)}$.*

Proof. By Lemma 13.5 and the crude count, the number of events with rough part r is at most $\min(2F_I(r), \tau^{*2}T/r)$, with τ^* as in §10. Hence

$$w_\ell \ll \log \log T \sum_{r \leq T, \ell \mid r} \frac{\min(2r^{\eta+o(1)}, \tau^{*2}T/r)}{r}.$$

Splitting at $R = T^{1/(1+\eta)}$ gives $w_\ell \ll R^\eta T^{o(1)}/\ell + \tau^{*2}T/(\ell R) = T^{\kappa+o(1)}/\ell$. With $z = T^{\kappa+\varepsilon}$ the hypothesis of Lemma 13.3 holds for large T , and

Lemma 13.3 with $S_{\text{tot}} = T^{o(1)}$ (Lemma 13.2) gives $\log(1/\delta^*) \leq \pi(z) \log T + T^{o(1)}$. $\square$

With $\eta = 3/5$ this gives $\kappa = 3/8$, weaker than Theorem 1.3; $\eta = 1/2$ gives exactly $1/3$. So individual Type I bounds beat the exponent $1/3$ only if $\eta < 1/2$, well beyond [4, Prop. 1.7]. The single-prime part of the per-prime hypothesis is not the bottleneck (Theorem 1.3 absorbs it by quarantining $T^{o(1)}$ bad primes); what is needed is control of $\sum_{r \leq T, \ell | r} F_I(r)/r$ uniformly in ℓ , a progression average of Type I counts (Assessment). The next theorem makes this route unnecessary.

Iterated quarantine. For an atom (M, D) and a set Π of primes let m_Π be the Π -part of M and $r_\Pi = M/m_\Pi$; the atom *survives* Π if $m_\Pi \mid 4D + 1$ and $r_\Pi > 1$. Put $S^* = \sum_{(M,D)} \max_\Pi \mathbf{1}[\text{survives } \Pi] / \varphi(r_\Pi)$.

Theorem 13.8 (Proved; the polylogarithmic form modulo [4, Prop. 1.4]).

$$\log(1/\delta^*(T)) \ll (\log T)^3 (S^* + 1).$$

Hence $\log(1/\delta^*(T)) \leq \exp(O(\log T / \log \log T)) = T^{o(1)}$, and, modulo [4, Prop. 1.4], $\log(1/\delta^*(T)) \ll (\log T)^7 / \log \log T$.

Proof. If (M, D) survives Π then $m_\Pi \mid g = \gcd(M, 4D + 1)$, so $1/\varphi(r_\Pi) \ll \log \log T \cdot g/M$; the proof of Lemma 3.3 with $m = g$ (the involution preserves g , $g \mid r' + k$, $g \mid 4sr'^2 + 1$) bounds S^* as in Lemma 13.2, giving $\exp(O(\log T / \log \log T))$, and $\ll (\log T)^4 \log \log T$ by [4, Prop. 1.4]. Let $z = (\log T)^2$, $k = \lfloor \log T / \log z \rfloor$, $c_0 = 1/(8k)$, $\Pi_0 = \{\ell \leq z\}$, and let $w_\ell(\Pi)$ be the Haar mass of the distinct events surviving Π whose rough part is divisible by ℓ . Repeatedly add to Π every prime $\ell \notin \Pi$ with $w_\ell(\Pi) > c_0$. If ℓ is added at some stage, c_0 is less than the sum of the weights $1/\varphi(r_\Pi)$ of atoms with $\ell \mid r_\Pi$; each atom has at most k primes above z and weight at most its contribution to S^* , so at most kS^*/c_0 primes are added. At the end every free prime has $w_\ell \leq c_0$, every event has at most k free primes, and the local lemma with $x_E = 2P(E)$ applies ($\sum_{E' \sim E} x_{E'} \leq 2kc_0 = 1/4$), giving probability $\geq \exp(-4S^*)$ of no event. The class of one modulo the quarantined prime powers costs $\leq (\pi(z) + 8k^2 S^*) \log T$. $\square$

This improves Theorem 1.3, and it makes the per-prime hypothesis of Lemma 13.3 and of Remark 13.4 unnecessary. On the prime side it supplies the per-prime conditions of Theorem 9.2, at the cost of $T^{o(1)}$ further quarantined primes. Below $\theta = 1/3$ the remaining obstacle there is the control of pair codegrees (Assessment). For $T \leq 10^6$ the iteration adds a few hundred primes in a single round, and the resulting certificates give $\log(1/\delta^*(10^6)) \leq 3.6 \cdot 10^3$ (EVIDENCE).

14. SUMMARY OF STATUS

- Proved: Lemmas 2.1–3.3, 7.1–8.2, 10.1, 12.1–12.3, 13.1, 13.2, 13.3, 13.5; Lemma 9.1 (classical); Theorem 9.2; Proposition 10.2, i.e.

$H_{\min}(\theta)$ for $\theta \geq 1/3$; Propositions 11.1, 11.3, 11.6, 12.4; Corollary 12.5; Theorem 1.3 (with the classical local lemma and divisor bound).

- Proved modulo Thorner–Zaman [15] (with the McCurley statement quoted there): Lemma 4.2, Theorems 5.1, 1.1, 6.1, 11.5, Corollary 12.7; hence the falsity of “ $W(p) \leq (\log p)^A$ for all large p ” for every $A < 3$.
- Proved modulo Lau–Wu [8, Prop. 5.1]: Theorem 12.6.
- Proved modulo Chang [2, Cor. 11]: the $5/8$ record (Remark 11.2).
- Proved: Theorem 13.8 ($\log(1/\delta^*) \leq T^{o(1)}$); its polylogarithmic form modulo [4, Prop. 1.4].
- Proved modulo the proof of [4, Prop. 1.7]: $F_I(n) \leq n^{3/5+o(1)}$. Proved implication: Theorem 13.7.
- Conditional on the per-prime hypothesis, and proved modulo [4, Prop. 1.4]: the polylogarithmic Haar bound of Remark 13.4.
- Open: $H_{\min}(\theta)$ for $\theta < 1/3$; an unconditional polylogarithmic Haar bound; any bound $W(p) > (\log p)^{3+\delta}$ or $ck_{\min}(p) > (\log p)^{1+\delta}$ infinitely often.

Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ or $ck_{\min}(p)$ may well have other representations.

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